Atom Physics

Lecture 2

October 24, 2026

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Course Contents

Lecture	Topics	Material
1	•Relativistic dynamics •Review electromagnetic waves •Photoelectric effect •Blackbody radiation	Book: Ch 2.1-2.5, 2.7, 3.1-3.3
2	**•Compton scattering** **•De Broglie's hypothesis** **•Experimental evidence for De Broglie waves** **•Uncertainty relationships for classical particles** **•Heisenberg uncertainty relationships**	**Book: Ch. 3.4, 4.1-4.4**
3	•Behaviour of a wave at a boundary •Confining a particle •The Schrödinger equation •Applications of the Schrödinger equation	Book: Ch. 5.1-5.4
4	•Basic properties of atoms •Scattering experiments and the Thomson model •The Rutherford nuclear atom •Line spectra •The Bohr model •The Franck-Hertz experiment •Deficiencies of the Bohr model	Book: Ch. 6.1-6.6, 6.8

Course Contents

Lecture	Topics	Material
5	•A one-dimensional atom •Angular momentum in the hydrogen atom •The hydrogen atom wave functions •Radial probability densities •Angular probability densities	Book: Ch. 7.1-7.7
6	•The Pauli exclusion principle •Electron states in many-electron atoms •Outer electrons: screening and optical transitions •Inner electrons: absorption edges and X-rays	Book: Ch. 8.1-8.3, 8.5
7	•Nuclear constituents •Nuclear sizes and shapes •Alpha decay •Beta decay •Gamma decay and nuclear excited states	Book: Ch. 12.1-12.3 12.6-12.9
8	Visit to the Catharina hospital, radiotherapy department	
Final Exam: Thursday October 30th, 13:30-16:30		

Recap Previous Lecture

- Introduction to Modern Physics
- Special Theory of Relativity

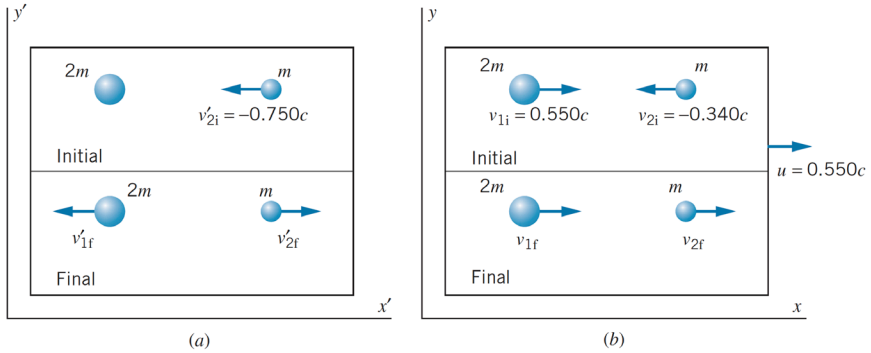
$$L = L_0\sqrt{1 - u^2/c^2}$$

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}$$

- Relativistic Dynamics

$$K = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

$$\vec{\boldsymbol{p}} = \frac{m\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}}$$

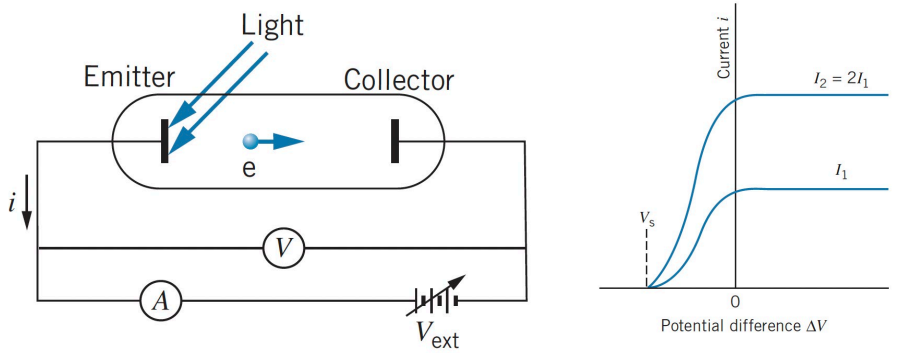


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Recap Previous Lecture

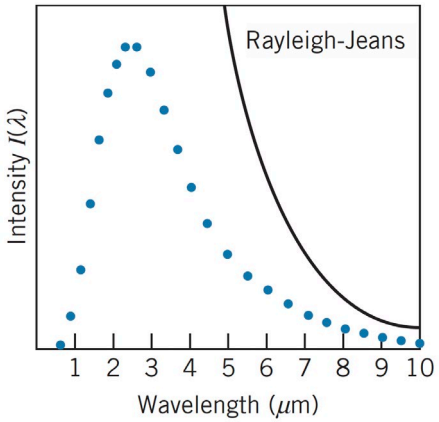
- Experiments showing that classical (wave) theory is not sufficient:

1. Photoelectric effect



$$K_{max} = eV_s = hf - \phi$$

2. Thermal Radiation



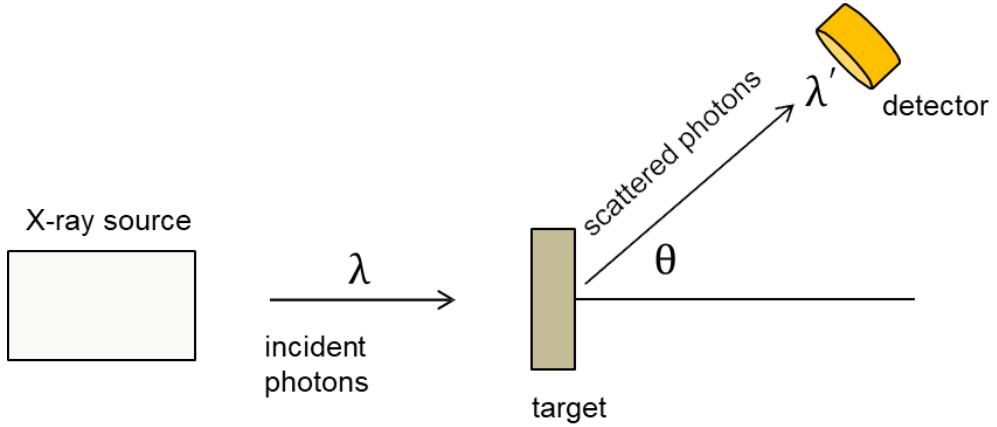
Planck's Radiation Law

$$I(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

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Compton Scattering

- Radiation that is scattering from loosely bound electrons



- Classical wave theory:
  - scattered radiation is less energetic…
  - But has the same wavelength as the incoming radiation

- Compton (1923): different wavelength at the detector λ'

Compton Scattering

- Using Einstein's Photon theory, x-rays can be described as photons

$$E = pc = hf$$

- Treat this problem as a collision

$$K = E + E_0 = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

$$\vec{\boldsymbol{p}} = \frac{m\vec{v}^2}{\sqrt{1-v^2/c^2}}$$

$$E = \sqrt{(pc)^2 + (mc^2)^2}$$

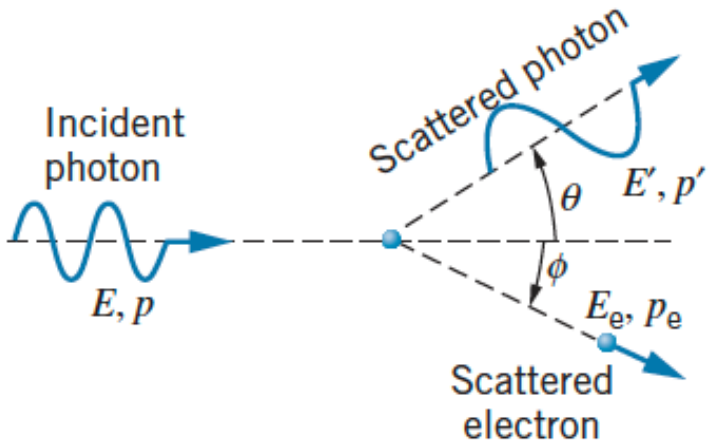
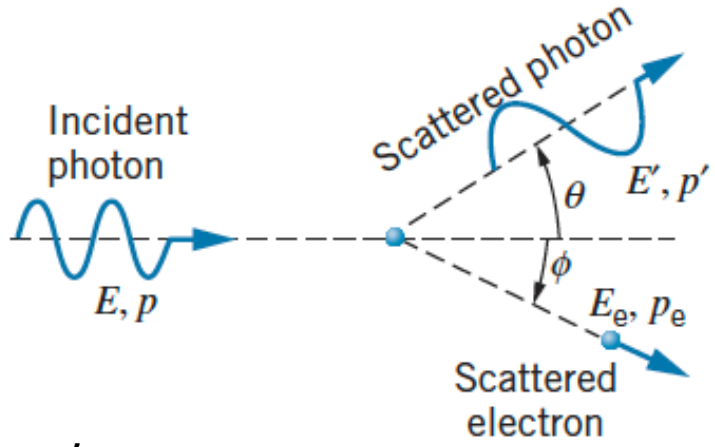


FIGURE 3.18 The geometry of Compton scattering.

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Compton Scattering



Before:

Electron is at rest $\vec{v} = 0$, photon is along the x-axis

$$E_p = pc$$
$$E_e = \sqrt{(p_e c)^2 + (m_e c^2)^2} = \sqrt{0 + (m_e c^2)} = m_e c^2$$

After (indicated with $'$):

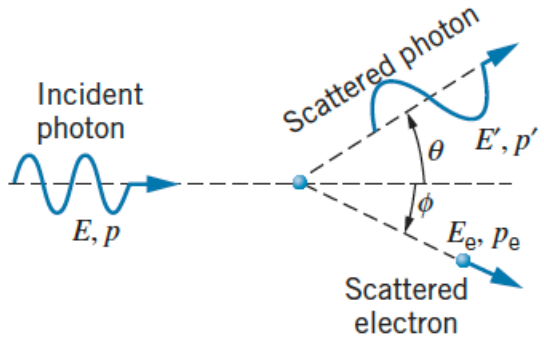
Both the electron and photon have energy and momentum:

$$E_p' = p'c$$
$$E_e' = \sqrt{(p_e' c)^2 + (m_e c^2)^2}$$

Conservation of energy:	$E_p + E_e = E_p' + E_e'$	(1)
Conservation of momentum (x-comp):	$p_p = p_p' \cos\theta + p_e' \cos\phi$	(2)
Conservation of momentum (y-comp):	$0 = p_p' \sin\theta + p_e' \sin\phi$	(3)

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Compton Scattering



Conservation of energy:	$E_p + E_e = E_p' + E_e'$	(1)
Conservation of momentum (x-comp):	$p_p = p_p' \cos\theta + p_e' \cos\phi$	(2)
Conservation of momentum (y-comp):	$0 = p_p' \sin\theta + p_e' \sin\phi$	(3)

From (2) & (3): $p_e'^{\,2} = p_p^2 - 2p_p p_p' \cos\theta + p_p'^{\,2}$

From (1): $E_p + m_e c^2 = E_p' + \sqrt{(p_e' c)^2 + (m_e c^2)^2}$

$$\left(E_p + m_e c^2 - E_p'\right)^2 = (p_e' c)^2 + \left(m_e c^2\right)^2$$

Substitute for $p_e'^{\,2}$ and dropping p-subscript :

$$\left(E + m_e c^2 - E'\right)^2 = c^2(p^2 - 2pp'\cos\theta + p'^2) + \left(m_e c^2\right)^2$$

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2}(1 - \cos\theta)$$

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2}(1 - \cos\theta)$$

Using $E = hf = \frac{hc}{\lambda}$:

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

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Compton Scattering

Arthur H. Compton
1892-1962



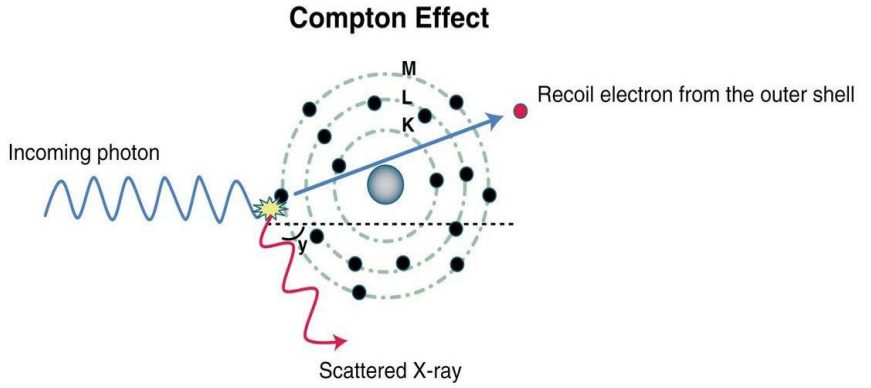
Compton Shift

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

- $\frac{h}{m_e c}$ is called the Compton wavelength

Note: Compton wavelength is not an actual wavelength, but a constant with dimension [m] that denotes maximal wavelength shift

- The Compton shift depends on the angle θ



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Compton Scattering – Experiment

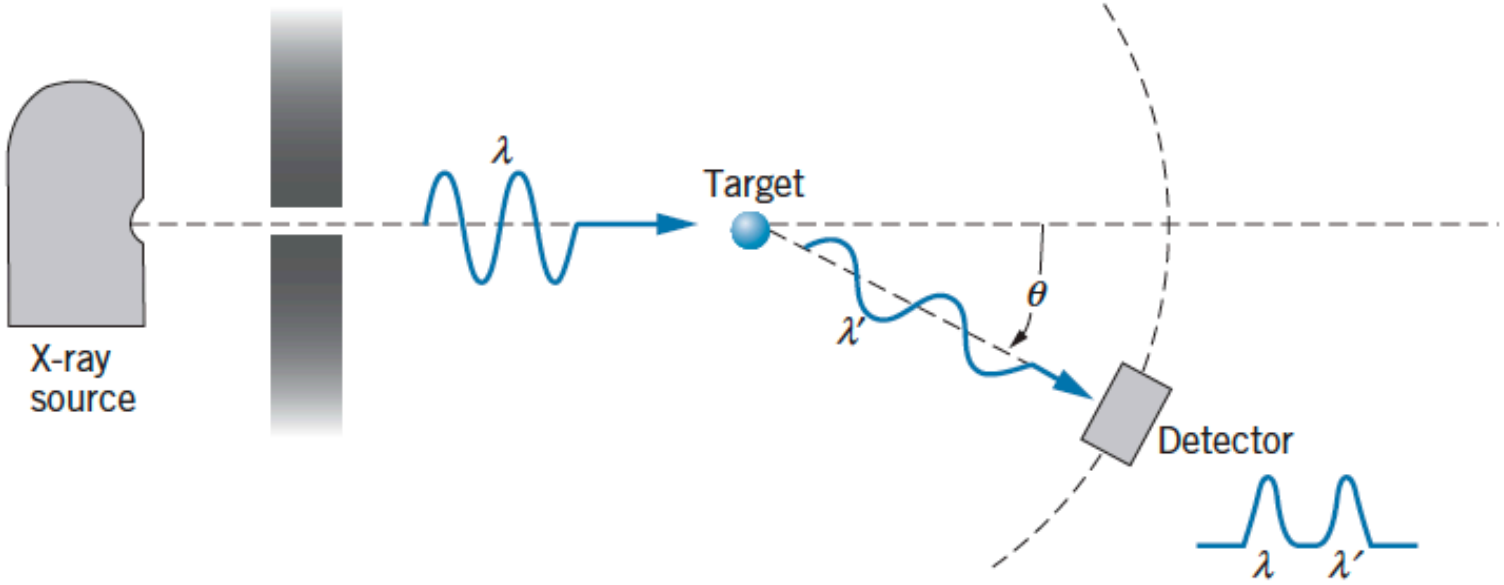


FIGURE 3.19 Schematic diagram of Compton-scattering apparatus. The wavelength λ' of the scattered X rays is measured by the detector, which can be moved to different positions θ. The wavelength difference $\lambda' - \lambda$ varies with θ.

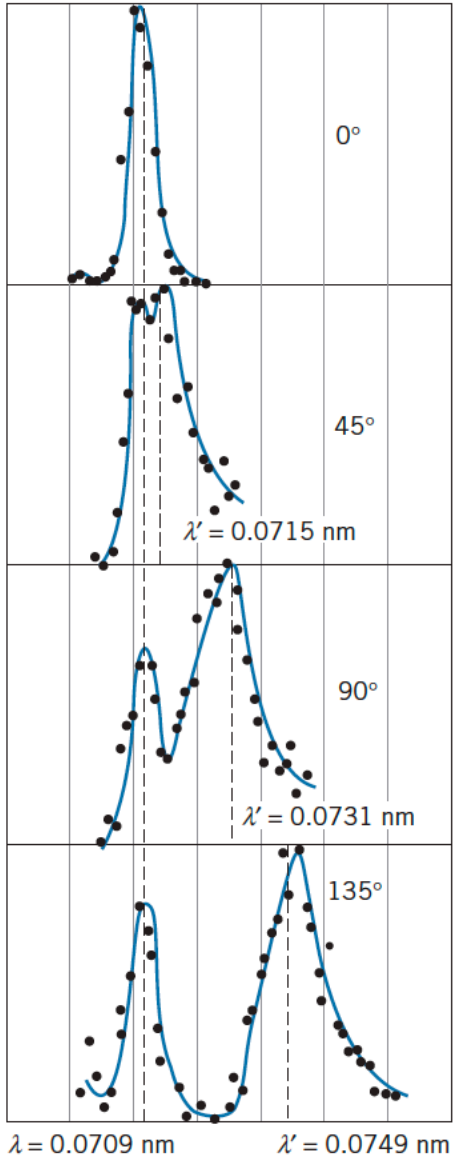


FIGURE 3.20 Compton's original results for X-ray scattering.

Compton Scattering – Exercise

X-rays with a wavelength of 0.24 nm are scattered by an angle of 40 degrees when passing a piece of carbon.

(a) What is the wavelength of the scattered X rays?

(b) What is the kinetic energy of the recoil electron?

$h = 6.626\ 10^{-34}$ Js
$m_e = 9.1094\ 10^{-31}$kg
$hc = 1240$ eV nm

Compton: $\lambda' - \lambda = \left(\frac{h}{m_e c}\right)(1 - \cos\theta)$

Photon energy: $E = hf = hc/\lambda = pc$

Electron energy: $K_e = \Delta E_{photon}$

Compton Scattering – Exercise

X-rays with a wavelength of 0.24 nm are scattered by an angle of 40 degrees when passing a piece of carbon.

(a) $$\lambda' - \lambda = \left(\frac{h}{m_e c}\right)(1 - \cos\theta) = \frac{6.626\ 10^{-34}\ \text{Js}}{(9.1094 \cdot 10^{-31}\text{kg}) \cdot (3.00 \cdot 10^{8}\text{ms}^{-1})}(1 - \cos 40°)$$

$$\Delta\lambda = (0.002426\ nm)(1 - 0.7660) = 0.0005676\ nm$$

So: $\lambda' = \lambda + \Delta\lambda = 0.24 + 0.0005676\ nm = \mathbf{0.2406\ nm}$

(b) $$K = E_{initial} - E_{scattered} \quad \text{and} \quad E_{photon} = \frac{hc}{\lambda}$$

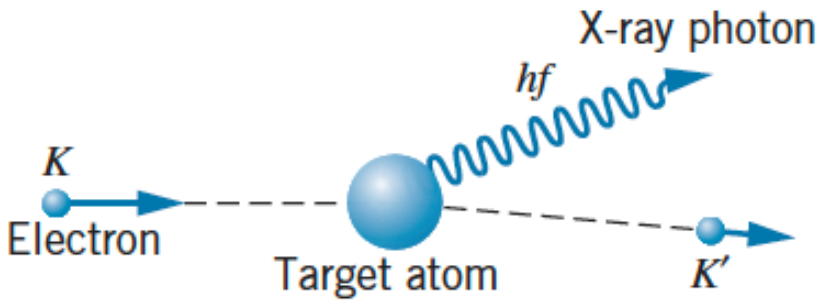
$$E_{initial} = \frac{1240\ eV\ nm}{0.24\ nm} = 5.167\ keV \quad , E_{scattered} = \frac{1240\ eV\ nm}{0.2406\ nm} = 5.154\ keV$$

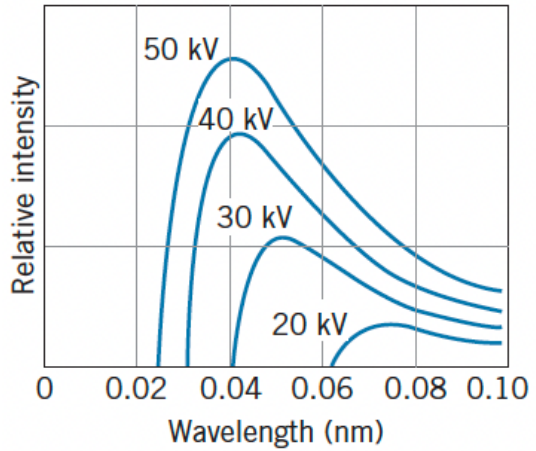
$$K = E_{initial} - E_{scattered} = \mathbf{13\ eV}$$

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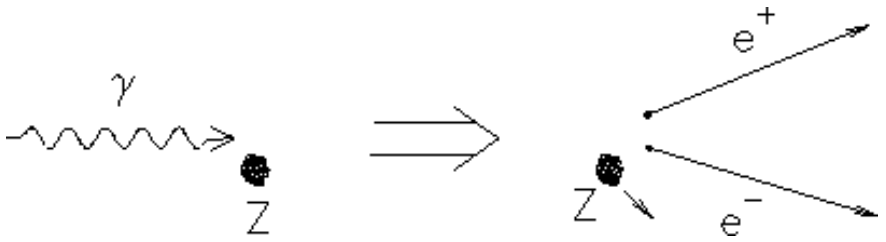
Other Interactions

- Bremsstrahlung

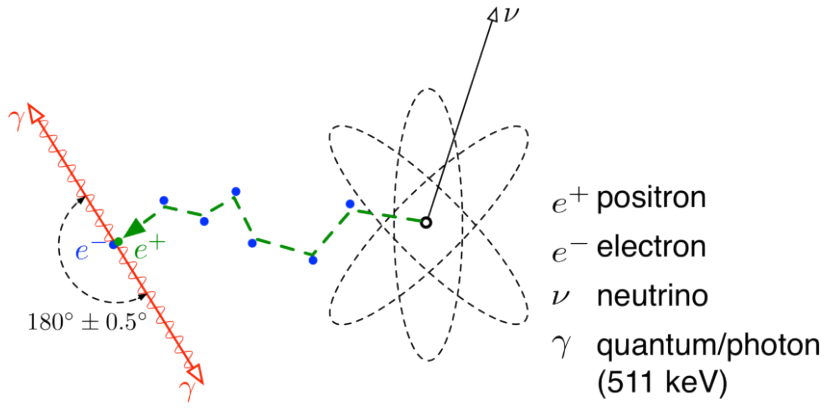


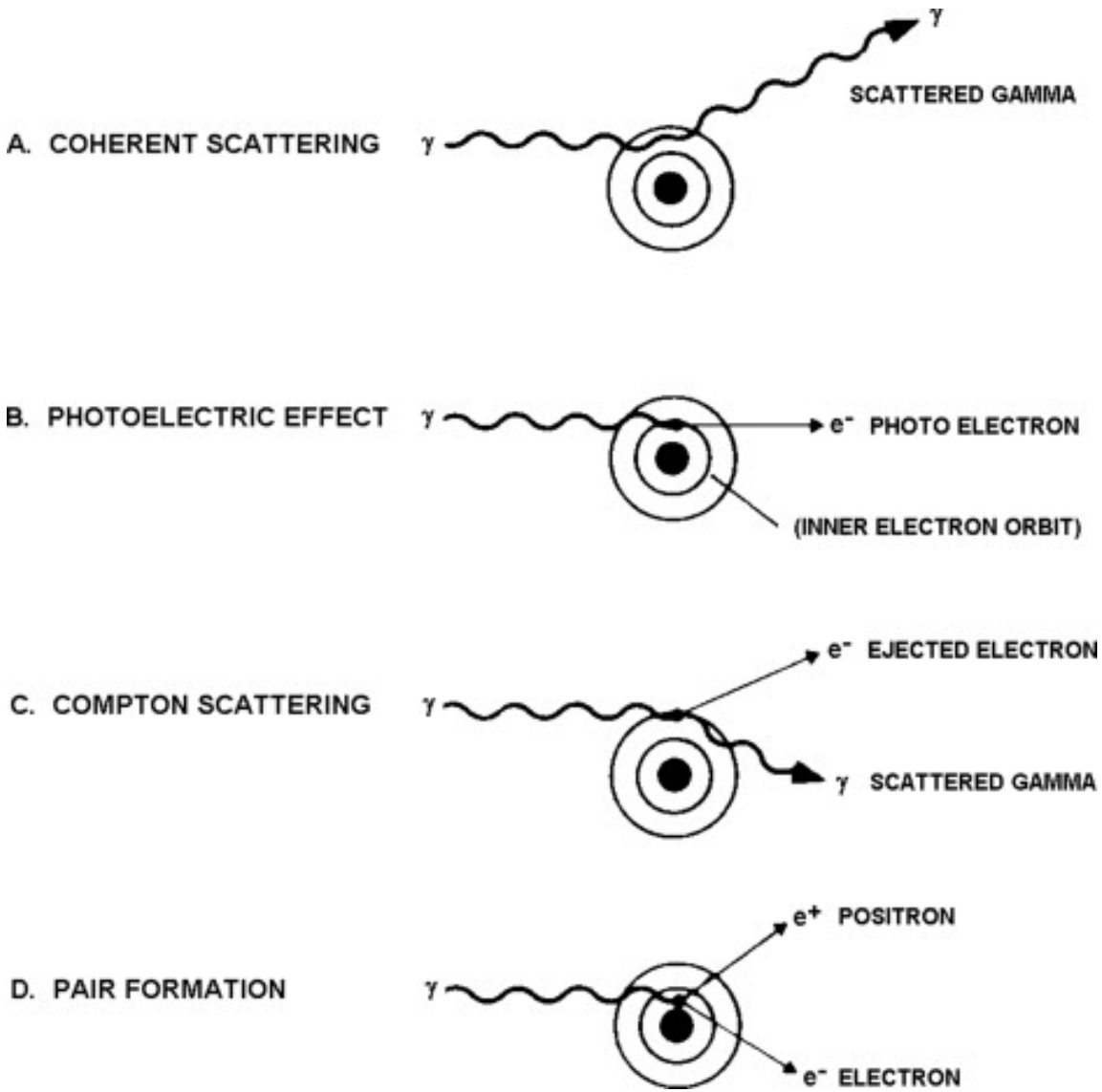


- Pair production (E>2*511 keV)



- Positron-electron annihilation




A. COHERENT SCATTERING
γ
γ
SCATTERED GAMMA
B. PHOTOELECTRIC EFFECT
γ
e⁻ PHOTO ELECTRON
(INNER ELECTRON ORBIT)
C. COMPTON SCATTERING
γ
e⁻ EJECTED ELECTRON
γ SCATTERED GAMMA
D. PAIR FORMATION
γ
e⁺ POSITRON
e⁻ ELECTRON

Medical Example

- Radiation Therapy:

Bremsstrahlung → photon beam formation

Compton scattering → interaction with tissue

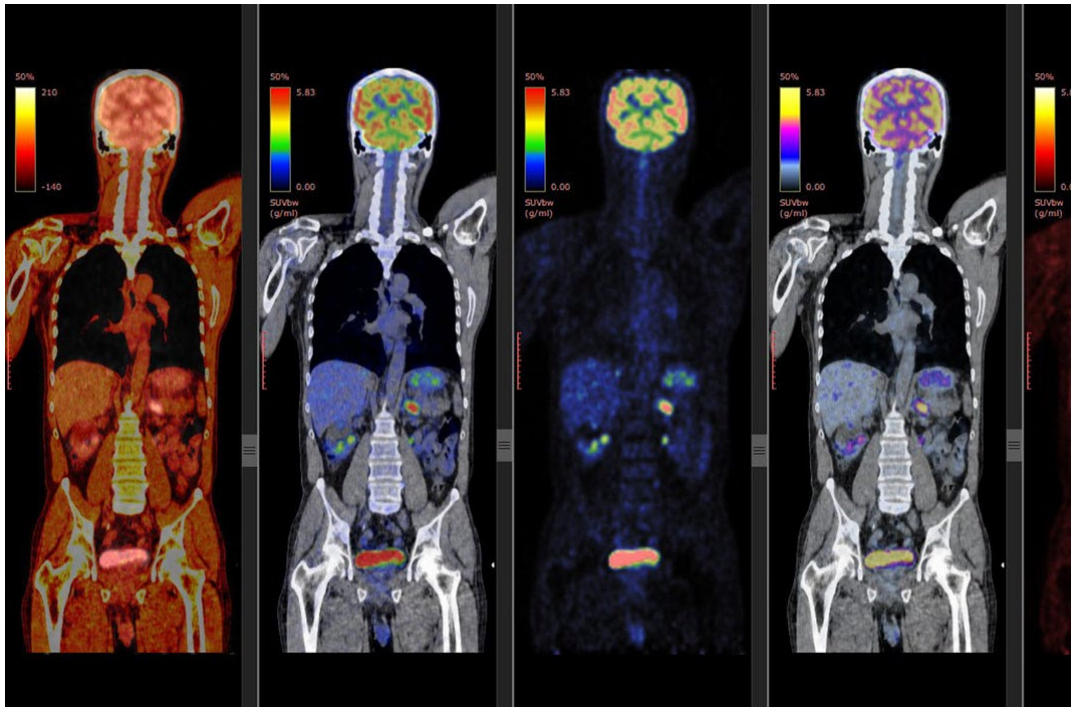


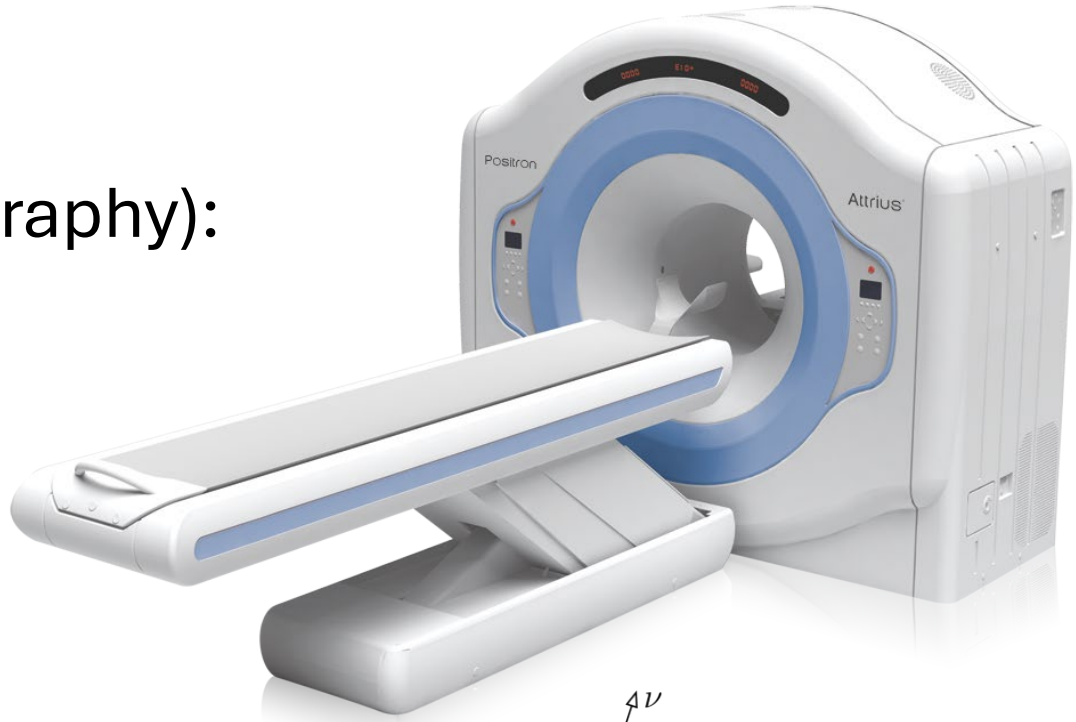
Image from: https://www.catharinaziekenhuis.nl/afdelingen/radiotherapie/

Medical Example

- PET-Scan (Positron Emission Tomography):

Positron-electron annihilation





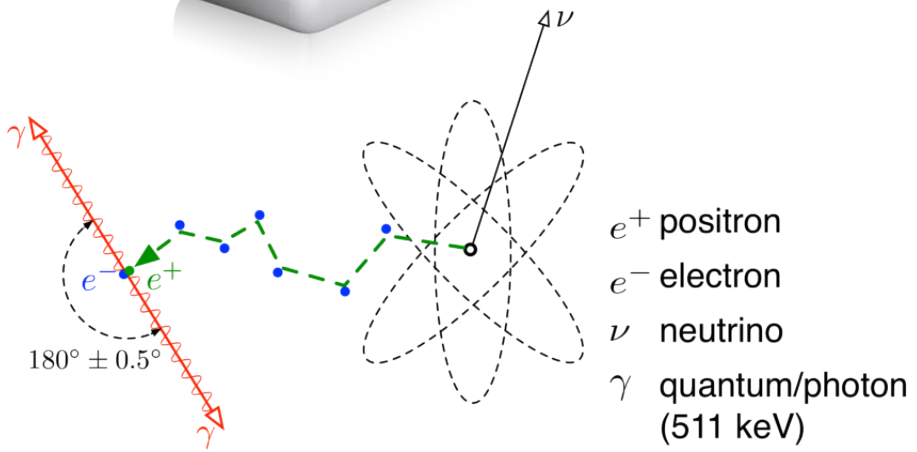
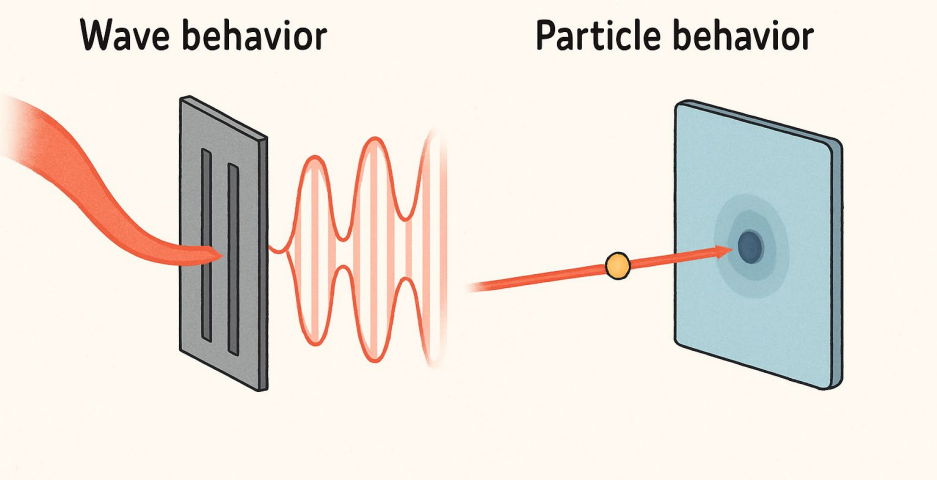


Image from: https://www.premierradiology.com/premier-radiology-tennessee-pet-ct/

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Wave-particle Duality

- **Photons** display both particle-like properties (e.g. momentum) and wave-like properties: wave –particle duality



- What about **particles**? Do they exhibit wave-like properties?

De Broglie Wavelength

Louis-Victor de Broglie
1892-1987



- De Broglie in 1924: If light has wave-like behaviour but also particle-like behaviour (photons), then the same should hold for particles:

- For photons: $E = hf = pc \rightarrow p = \frac{h}{\lambda}$

- For particles: $\boxed{\lambda = \frac{h}{p} = \frac{h}{mv}}$ → De Broglie wavelength

De Broglie Wavelength – Exercise

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Compute the de Broglie wavelength of the following

(a) 1000-kg automobile traveling at 100 m/s

(b) 10-g bullet traveling at 500 m/s

(c) A smoke particle of mass 10−9 g moving at 1 cm/s.

(d) An electron with a kinetic energy of 1 eV

(e) An electron with a kinetic energy of 100 MeV

De Broglie Wavelength – Exercise

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Compute the de Broglie wavelength of the following

(a) 1000-kg automobile traveling at 100 m/s → $6.6 \cdot 10^{-39}$ m

(b) 10-g bullet traveling at 500 m/s → $1.3 \cdot 10^{-24}$ m

(c) A smoke particle of mass 10–9 g moving at 1 cm/s. → $6.6 \cdot 10^{-20}$ m

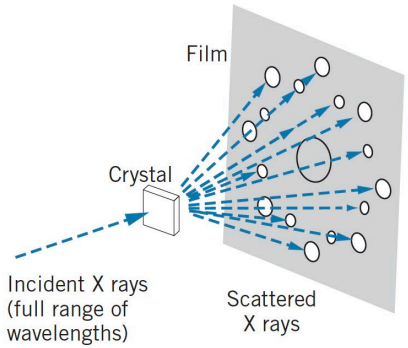
(d) An electron with a kinetic energy of 1 eV → $1.2 \cdot 10^{-9}$ m

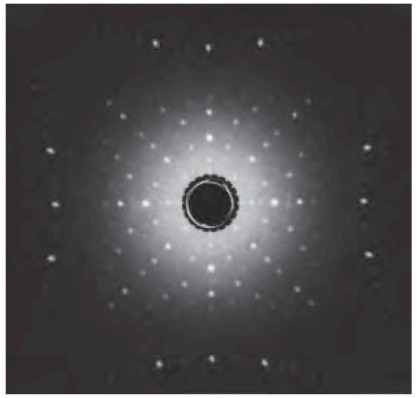
(e) An electron with a kinetic energy of 100 MeV → $1.2 \cdot 10^{-14}$ m

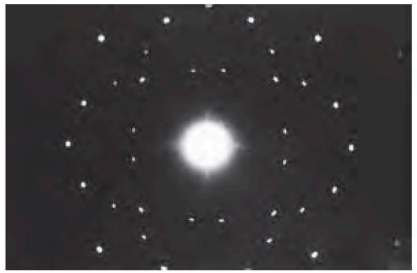
Because of the smallness of h, only for particles of atomic or nuclear size will the wave behavior be observable!

Diffraction Experiment

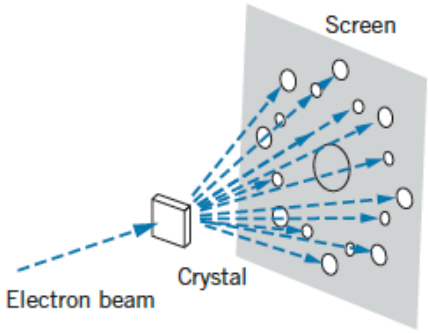
- Remember from previous lecture
  Using X-rays (waves):

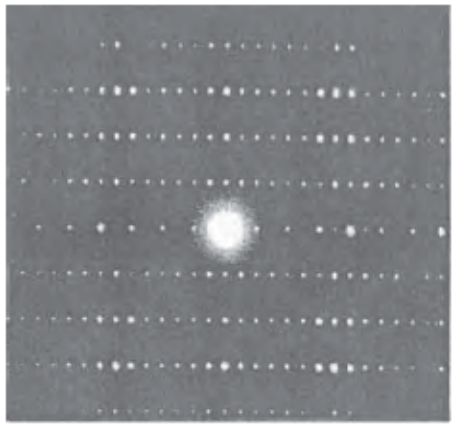






- Now, using electrons (particles):

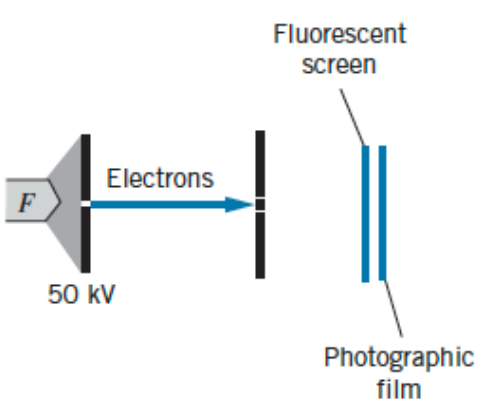


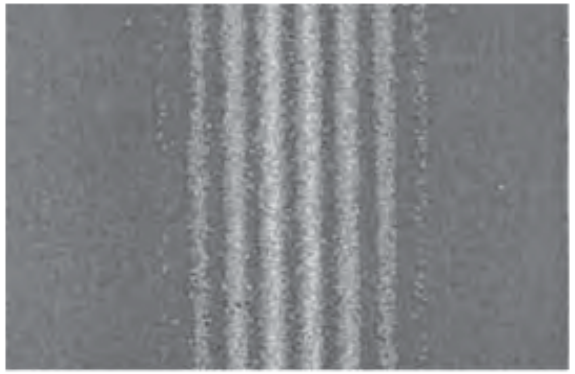


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Double Slit Experiment

- Double slit experiment with particles:





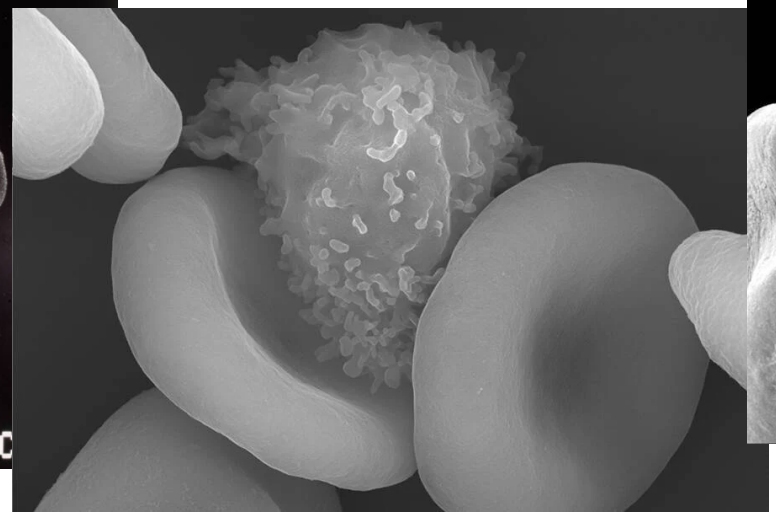
- These experiments show the wave character of particles (electrons, neutrons, protons, atoms, and molecules)

Application – Electron Microscope

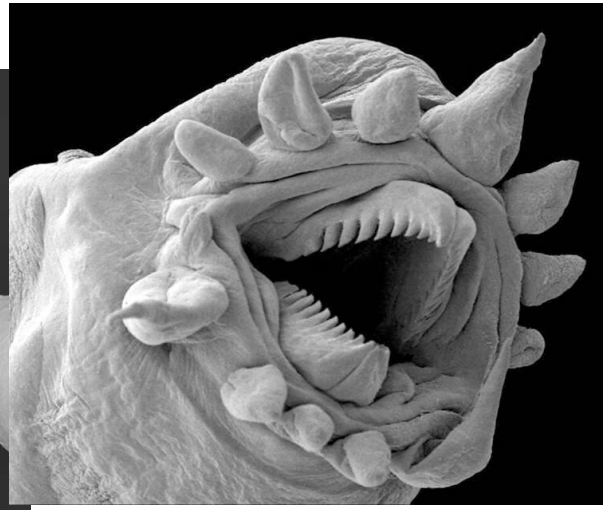
- The electron microscope uses electron waves to illuminate and form an image of objects
- Electromagnets function as lenses
- Resolution of around $0.1\ \mathrm{nm}$ (resolution for light microscopes $\sim 200\ \mathrm{nm}$)



The tip of a screw worm fly larva.



Human red blood cells and a lymphocyte.

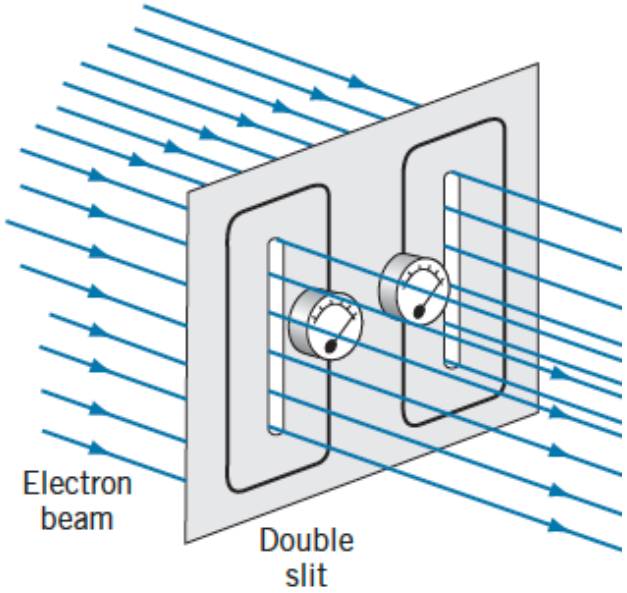


Closeup on the mouth of a hydrothermal worm. Image width $568\ \mu m$

https://allthatsinteresting.com/electron-microscope-images

Wave-Particle Duality

- What if we try to determine through which slit the particles are going?



Wave-Particle Duality



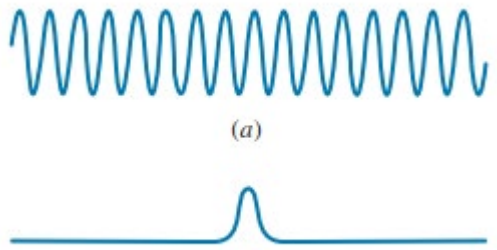
Delayed choice double slit experiment



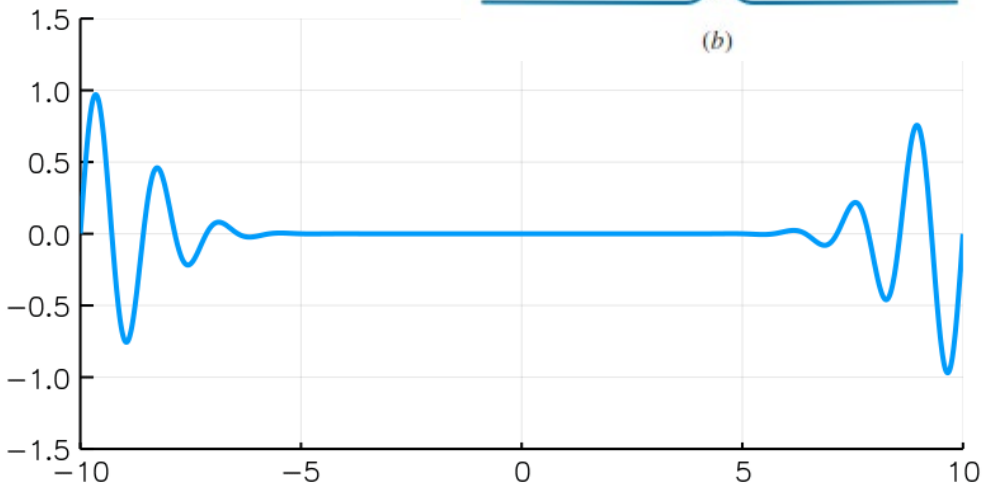
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Wavy particles?

- Classical *Particles* are located in a distinct place
- Classical *Waves* are everywhere



- *Photons, electrons, etc*
  are neither & both: ‘wave packets’



By Becarlson - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=67433782

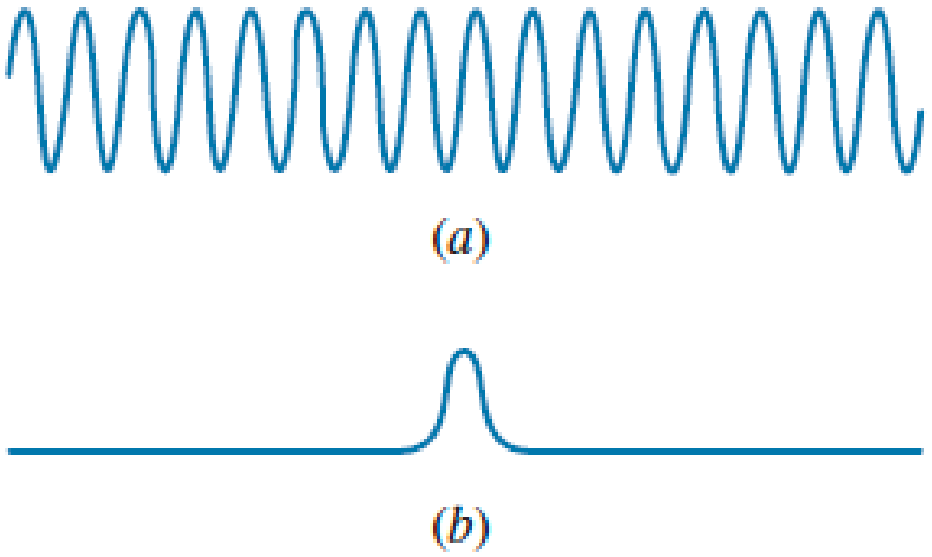
- But how to determine position and frequency?

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Uncertainty Relationships for Waves

Have a look at both waves

1) How would you determine the position of the wave?
2) How would you determine the wavelength?



Uncertainty Relationships for Waves

Error in wavelength:

$$\Delta\lambda \sim \epsilon\lambda \quad \text{with } \epsilon \sim 0.1$$

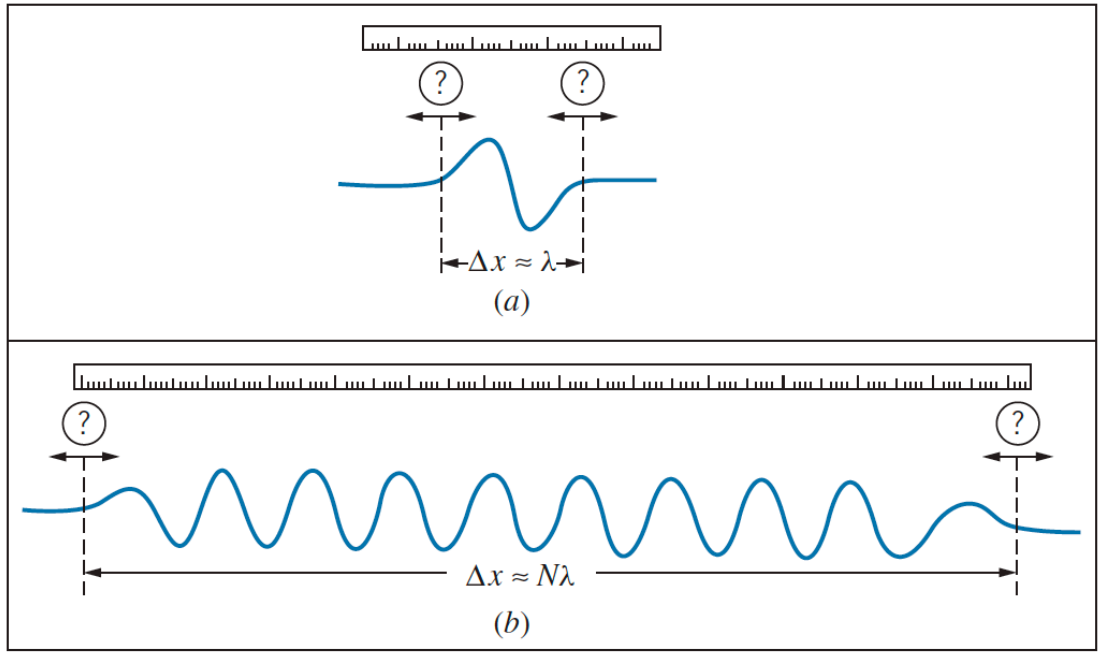
Error in position is about the size of the packet: $\Delta x \approx \lambda$
Total error/uncertainty:

$$\Delta x \Delta\lambda \sim \epsilon\lambda^2$$

Small packet: small error in Δx, but large uncertainty in $\Delta\lambda$

Large packet with N periods: $\Delta\lambda = \varepsilon\lambda/N$. But at the expense of increasing $\Delta x = N\lambda$;

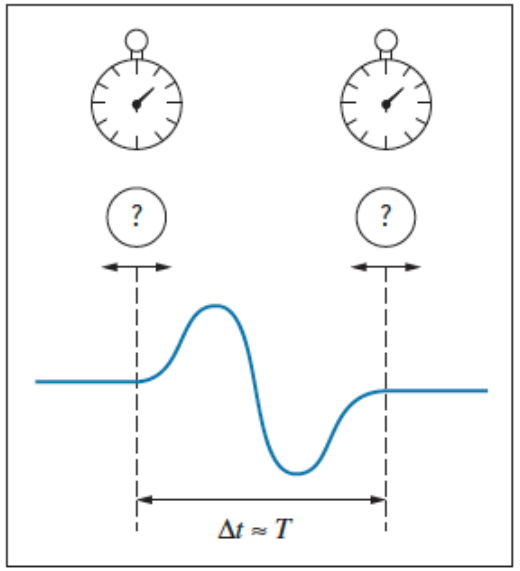
$$\Delta x \, \Delta\lambda \sim (N\lambda)(\varepsilon\lambda/N) = \varepsilon\lambda^2$$



Uncertainty Relationships for Waves

Uncertainty localization / wavelength: $\Delta x \Delta\lambda \sim \epsilon\lambda^2$

Uncertainty duration / period of wave: $\Delta t \Delta T \sim \epsilon T^2$



In terms of frequency: $f = \frac{1}{T}$ Note: $\Delta f \neq \frac{1}{\Delta T}$!

$\frac{df}{dT} = -\frac{1}{T^2} \Rightarrow \Delta f = \frac{1}{T^2}\,\Delta T$ (only interested in magnitude, ignore minus sign)

Uncertainty time/frequency of wave: $\Delta f \Delta t \sim \epsilon$

TU/e

Uncertainty Relationships for Waves

- The uncertainty principle applies to *any* wave, also De Broglie waves:

  De Broglie: $p = \frac{h}{\lambda} \Rightarrow \frac{dp}{d\lambda} = -\frac{h}{\lambda^2} \Rightarrow \Delta p = \frac{h}{\lambda^2} \Delta\lambda$

  Therefore: $\Delta x \Delta\lambda \sim \epsilon\lambda^2 \Longrightarrow \Delta x \Delta p \sim \epsilon h$

- According to quantum mechanical calculations: $\epsilon_{min} = \frac{\pi}{4}$

$$\Delta x \Delta p \geq \frac{1}{2}\hbar \text{ with } \quad \hbar := \frac{h}{2\pi}$$

- In practice we do worse than this limit, so we can take: $\Delta x \Delta p \sim \hbar$

Heisenberg Uncertainty Principles

Werner Heisenberg
1901-1976



1) “It is impossible to determine simultaneously the position and momentum of a particle with arbitrary precision”

$$\Delta x \Delta p_x \geq \hbar$$

also for y, z

2) “It is impossible to determine simultaneously the energy and time coordinate of a particle with arbitrary precision”

$$\Delta E \Delta t \geq \hbar$$

Heisenberg Uncertainty Principles – Exercise

- An electron moves with a velocity of $3.6 \cdot 10^6$ m/s. This velocity is measured with a precision of 1%. What is the maximal precision of a measurement of the position of the electron?

$$m_e = 9.11 \cdot 10^{-31}\text{kg}$$

- A ball moves with a velocity of 40 m/s. This velocity is measured with a precision of 1%. What is the maximal precision of a measurement of the position of the ball?

$$m_{ball} = 0.15\ \text{kg}$$

Heisenberg Uncertainty Principles – Exercise

- An electron moves with a velocity of $3.6 \cdot 10^6$ m/s. This velocity is measured with a precision of 1%. What is the maximal precision of a measurement of the position of the electron?

$$m_e = 9.11 \cdot 10^{-31}\text{kg}$$

$$p_x = mv_x = \left(9.11 \cdot 10^{-31}\right)\left(3.6 \cdot 10^6\right) = 3.3 \cdot 10^{-24}\ \text{kg}\frac{\text{m}}{\text{s}}$$

$$\Delta x \sim \frac{\hbar}{\Delta p_x} = \frac{\left(1.05 \cdot 10^{-34}\right)\text{Js}}{(3.3 \cdot 10^{-24} * 0.01)\text{kg m/s}} = 3.2\ \text{nm}$$

- A ball moves with a velocity of 40 m/s. This velocity is measured with a precision of 1%. What is the maximal precision of a measurement of the position of the ball?

$$m_{ball} = 0.15\ \text{kg}$$

$$p_x = mv_x = (0.15)(40) = 6\ \text{kg m/s}$$

$$\Delta x \sim \frac{\hbar}{\Delta p_x} = \frac{\left(1.05 \cdot 10^{-34}\right)\text{Js}}{(6 * 0.01)\text{kg m/s}} = 1.75 \cdot 10^{-33}\ \text{m}$$

HEISENBERG UNCERTAINTY RELATIONSHIPS HAVE NO MACROSCOPIC CONSEQUENCES

Wave Packets

- Particles can be represented by wave packets

- To localize a particle, we need to combine many sine waves (Fourier superposition) → the result is a **wave packet**.

- Uncertainty principle implies that no two measurements will yield identical results (‘throwing a die’) → we need statistics to describe generalized outcome of many experiments

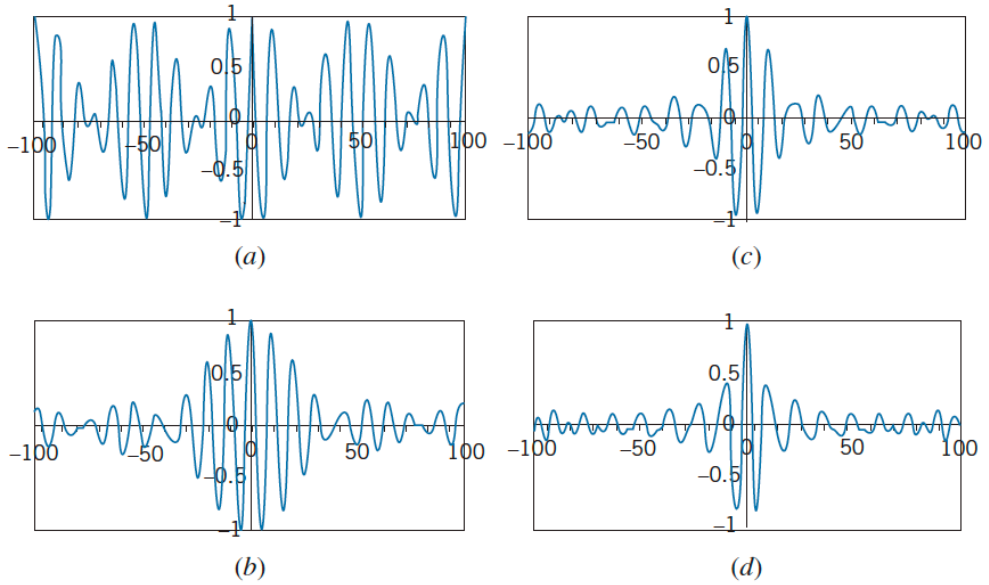
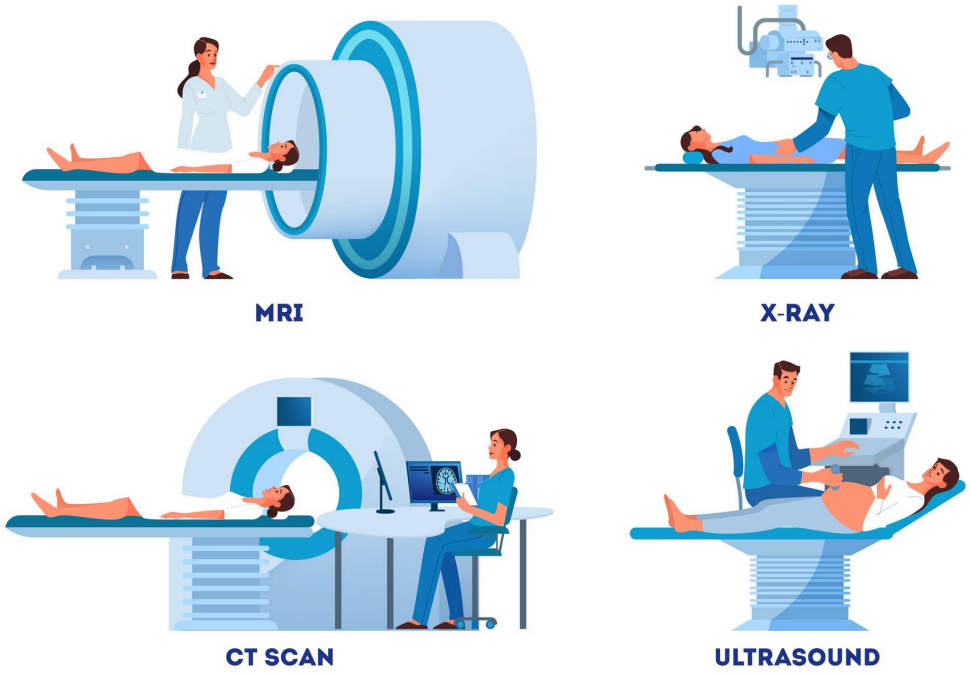


FIGURE 4.23 (*a*) Adding two waves of wavelengths 9 and 11 gives beats. (*b*) Adding 5 waves with wavelengths ranging from 9 to 11. (*c*) Adding 9 waves with wavelengths ranging from 8 to 12. (*d*) Adding 13 waves with wavelengths from 7 to 13. All of the patterns repeat from $-\infty$ to $+\infty$.

We will see more about these probability calculations in the next lecture!

Why Do We Care?

- These principles are not “just theory”. They **define the boundaries** of what is possible in medical imaging, radiation therapy, and radiation safety.





Limit time spent near a radiation source, increase distance away from a radiation source, or use shielding.

Images from: https://www.pihhealth.org/wellness/articles/understanding-the-differences-between-types-of-medical-imaging/ & https://www.versantphysics.com/2021/04/08/alara-the-gold-standard-of-radiation-protection/